

ARNAV MALHOTRA

Student Researcher & Engineer

am@welcra.com | welcra.com | linkedin.com/in/welcra | github.com/welcra

RESEARCH & PATENTS

Tesseractarius | *Hardware Architecture*

Preprint & Patent in drafting

- A fundamental resolution to the $O(N)$ syndrome decoding latency bottleneck in fault-tolerant quantum error correction. Designed to bypass the catastrophic time-scaling of traditional decoders.
- The architecture is mathematically agnostic to underlying QECC topology (supporting qLDPC, honeycomb, and rotated surface codes). Currently undergoing physical synthesis and benchmarking. Technical specifications and physical claims are restricted pending USPTO filing.

Praecursor

Patent Pending

- TITLE: SYSTEM AND METHOD FOR DYNAMIC STATE TRANSPOSITION AND PREEMPTIVE EVASION OF SPATIOTEMPORAL CORRELATED ERRORS
- This filing presents a novel method for mitigating the impact of spatiotemporal correlated errors (nonrandom quantum noise coming from sources like cosmic rays and radioactivity) in quantum processors. Standard post hoc correction, while working well with random errors, suffers great difficulty in correcting correlated errors. This method involves detection of correlated errors before their impact is materialized and the transportation of the quantum information outside of the predicted path of error propagation, offering a step closer to fault-tolerant quantum computing.

fsynth: Synthetic Financial Universes

Python Package, Preprint

- Traditional machine learning models trained on financial market data are known to be highly susceptible to overfitting the past market. Historical data is limited, and models trained only on data from the past will only be able to perform well in financial conditions mimicking the past. To combat this data scarcity, many use mathematical generators of unlimited synthetic data. However, traditional synthetic data generators often rely on Geometric Brownian Motion (GBM), which fails to account for empirical phenomena such as volatility clustering, leverage effects, and "black swan" jump events.
- This paper introduces fsynth, a multi-factor synthetic engine that integrates a Heston-based stochastic volatility model with Merton jump diffusion, governed by a hidden Markov regime-switching process. By using macro-regime states alongside company specific "genes," the engine generates plausible universes containing both price action and fundamental accounting data. We demonstrate that while the engine can be calibrated to replicate conditions of historical benchmarks, its primary utility lies in its parameterizable nature, allowing for the simulation of arbitrary, "out-of-sample" stress scenarios.

Quantum Binomial Options Hedging Framework

Preprint

- The Quantum Binomial Options Pricing Model is a novel approach to pricing derivatives. It serves as a quantum analogue to the classical binomial options pricing model, using concepts from quantum mechanics like the Pauli matrices and Bloch sphere to represent the derivatives market. Through usage of a density matrix and mixed states, the model can potentially price derivatives more accurately. However, at this time there exist no clear formulae and methods for hedging derivatives through this model.
- This paper numerically tests the performance of the hedging framework corresponding to the model as compared to the classical model using a year of historical SP 500 data. The simulation reveals that the Quantum Binomial Options Pricing model is, at least in its current state, not viable for usage in actually hedging derivatives. This is due to numerical instabilities, particularly in the Quantum Theta and Quantum Vega, the former of which ends up negative, betraying fundamental financial postulates, and the latter of which is blown out of scale. We hypothesize that these nonsensical values for Quantum Theta and Quantum Vega stem from the single period nature of the model we tested, as well as the isotropic assumption of the operator A . We conclude that the single period Quantum Binomial Options Pricing Model is currently unsuitable for practical applications of hedging and that our findings reveal the necessity of rigorously testing new financial models against benchmark models like the Classical Binomial Options Pricing Model.

PROJECTS & OPEN SOURCE

yfinance Contributor | 30M+ Monthly Downloads

- Engineered fallback mechanisms for the industry-standard Python library for Yahoo! Finance data. Resolved API dropouts by implementing HTML-parsing redundancy for critical quantitative metrics.

EVE Online Market Arbitrage | *Commodity Analytics*

- Developed an intraregional arbitrage calculator and predictive commodity modeling system for the Jita market, analyzing synthetic macroeconomic behavior in massive multiplayer environments.

EXPERIENCE

Wharton Global High School Investment Competition

Global

Lead Quantitative Researcher

Winner (1/6300 Globally)

- Placed in the top 1 out of 6300+ teams globally. Engineered a multi-asset quantitative strategy utilizing an ensemble of Black-Litterman models, multi-objective optimization, and an ML-based crisis prediction framework for robust risk mitigation.

The Stuyvesant Spectator

New York, NY

Web Lead

- Maintain the digital infrastructure for an award-winning high school publication. Lead the web department in optimizing content delivery, preserving site reliability, and ensuring continuous uptime.

AI CDST

New York, NY

Software Member

- Working with a team under the mentorship of Dr. Priyank Jain to create a generative AI tool to aid nurses in rural parts of West Bengal, India.

EDUCATION

Stuyvesant High School

New York, NY

Class of 2028

- Coursework includes AP Calculus BC, AP Precalculus, AP Statistics, AP Environmental Science, and AP Human Geography.